FARMSKING TECHNOLOGIES PRIVATE LIMITED

Complete Project Structure Tree & Architectural Blueprint

CIN: U01100PB2026PTC063294
A4 Standard: 210mm × 297mm | ISO 9001 / 27001
Generated: November 8, 2026 | Verified Architecture

BACKEND MODULES	DATABASE SCHEMA	FRONTEND (USER APP)	ADMIN CONSOLE
60 Modules	**19 Schemas / 103 Tables**	**Expo Web & Mobile**	**Full Governance**
NestJS Modular Domain Services	PostgreSQL Multi-File Prisma	File-Based Router (Port: 8081)	Multi-Role Console (Port: 8082)

Corporate Identity & Statutory Compliance: Registered under Companies Act, 2013 (MCA). Registered Office: VPO Goindwal Sahib, Tehsil Khadur Sahib, Tarn Taran, Punjab - 143422.

PRODUCTION READY

1. Root Repository Structure

Repository Root (d:\farmsking)

```
d:armsking                          # Root Corporate Monorepo Workspace
├── backend/                        # NestJS Enterprise API Gateway (Port: 3000)
├── frontend/                       # Expo React Native Web Client (Farmers, Partners, Buyers - Port: 8081)
├── admin/                          # Expo React Native Web Admin Console (Executive Governance - Port: 8082)
├── Govt/                           # Confidential Statutory & Legal Dossiers (Guarded by .gitignore)
│   └── docs/                       # MCA Incorporation, e-MoA, e-AoA, PAN, TAN, GSTIN, MSME, Shareholding
├── docs/                           # Technical Architecture Documentation & ICAR Agronomy Specifications
├── reports/                        # Automated Statutory Reports, Invoices & Tax Audit Exports
├── PROJECT_RULES.md                # Master Corporate Guidelines & Architectural Invariants
├── AGENTS.md                       # AI Agent Behavioral Invariants (Pure English UI, Laptop Benchmark, Auto-Allow)
├── GEMINI.md                       # Gemini Engineering Rules, Strict A4 Rule, Technical Verification Invariants
├── package.json                    # Workspace Orchestration & Script Automation
└── tsconfig.base.json              # Shared TypeScript Compiler Configurations
```

2. Backend Architecture - NestJS & Prisma Multi-Schema Engine

backend/ (Port: 3000)

```
backend/
├── prisma/
│   ├── schema/                     # 19 Schema Files (Total 103 Tables with Timestamp Logging)
│   │   ├── schema.prisma           # Prisma Client Generator & PostgreSQL Datasource Config
│   │   ├── users_farms_plots_crops.prisma  # User (208 fields), Farm, Plot, CropCycle
│   │   ├── section_192.prisma      # SellerStore, SubOrder, Gstr8Report, SellerInvoice, Payouts
│   │   ├── section_378.prisma      # FarmerPlan, Upgrade Quotes, Prorated Credits, DoctorChangeRequest
│   │   ├── section_636.prisma      # KingConnectLink, DemandRequest, KingPaymentRequest, P2P Ledgers
│   │   ├── section_860.prisma      # AppSetting, AuditLog (Timestamped Tables), WalletTransaction, BankBeneficiary
│   │   ├── section_309.prisma      # TrainerAssignment, ISO Controls, Agronomy Expert Audits
│   │   ├── expenses_sales_customers_payments.prisma  # Expense, Party, CropSale, JFormReceipt, MandiPrice
│   │   ├── gardener_customer_business_partner_dashboards.prisma  # Product, Order, OrderItem, Review
│   │   ├── labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma  # Farm Resource Logs
│   │   ├── ai_doctor_crop_health_treatments.prisma  # AI Scans, Diagnostics, Prescriptions
│   │   ├── satellite_telemetry_weather_soil.prisma  # Satellite NDVI Scans, Soil Sensors, Weather Cache
│   │   └── banking_escrow_settlements_tax.prisma  # ICAR Protocols, GST/TCS Ledgers, Escrow Holds
│   └── migrations/                 # Version-Controlled SQL Database Migrations
├── src/
│   ├── main.ts                     # App Bootstrap, CORS, ValidationPipe, Swagger OpenAPI Docs (/api/docs)
│   ├── app.module.ts               # Root NestJS Module importing all 60 Domain Modules
│   ├── common/
│   │   ├── guards/                 # JwtAuthGuard, RolesGuard (SUPERADMIN, FARMER, SELLER, EXPERT, BUYER)
│   │   ├── interceptors/           # AuditLoggingInterceptor (Logs mutations with timestamps & diffs)
│   │   ├── decorators/             # @CurrentUser(), @Roles(), @Public()
│   │   └── filters/                # HttpExceptionFilter with Standard Error Serialization
│   └── modules/                    # 60 Modular NestJS Domain Services (Controllers, Services, DTOs)
```

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Backend Modules Categorization & Domain Breakdown (60 Modules)

CIN: U01100PB2026PTC063294
Architecture: NestJS Micro-Modular REST API
Status: 100% Type-Safe Verified Build

3. Backend 60 Modules Grouped by Business Functional Pillar

60 Total Modules

Pillar 1 Identity, Access & Multi-Role RBAC — 6 Modules

- **auth** • JWT Authentication, OTP verification, Password resets, Session lifecycle
- **users** • User profile management (208 fields), KYC status, King ID generation
- **roles** • Role-Based Access Control guards (SUPERADMIN, FARMER, SELLER, etc.)
- **kyc** • Aadhaar & PAN verification, Document uploads, Verification approval flows
- **referrals** • Multi-tier referral engine, Referral codes, Royalty distributions
- **customer-addresses** • Delivery addresses, Geolocation tagging, Village PIN routing

Pillar 2 Agronomy & Field Operations — 11 Modules

- **farms** • Multi-farm landholding registry, Geo-fenced boundaries, Soil profiles
- **plots** • Sub-plot partitioning, Soil chemistry, Irrigation infrastructure tracking
- **crop-cycles** • Sowing to harvest lifecycles, Variety tracking, Target yield benchmarks
- **crop-calendar** • Dynamic ICAR-aligned stage advisory, Automated activity reminders
- **expenses** • Crop expenditure logging, Input cost tracking, Cost-per-acre analysis
- **harvest** • Yield logging, Quality grading, Moisture content, Weighbridge records
- **irrigation** • Tube-well and canal water schedules, Soil moisture thresholds
- **labour** • Labour crew logging, Daily wage registers, Attendance tracking
- **spray** • Pesticide/herbicide spray registers, Dosage logging, Safety intervals
- **fertilizer** • NPK application logs, Organic compost records, Micronutrient telemetry
- **machinery** • Tractor, Harvester, Diesel usage, Implement maintenance logs

Pillar 3 AI Crop Doctor & Tele-Agri Advisory — 5 Modules

- **ai-doctor** • Multi-spectral leaf vision diagnosis, Real-time disease identification
- **crop-health** • Continuous stress index tracking, Pest outbreak alerts, Health scores
- **diagnostics** • Lab test reports, Soil sample results, Historical diagnostic records
- **experts** • Agronomist directory, Doctor availability, Video/Audio consultation scheduling
- **doctor-reviews** • Farmer ratings and feedback for agronomy advisors and trainers

Pillar 4 AgriStore Multi-Vendor E-Commerce — 8 Modules

- **products** • Seed, fertilizer, pesticide catalog, Stock inventory, Pricing
- **categories** • Hierarchical product categories, Department navigation, Filter facets
- **orders** • Master purchase orders, Multi-vendor cart checkout, UTR tracking
- **sub-orders** • Vendor-specific split orders, Individual dispatch status, Seller fulfillment
- **seller-store** • Vendor storefront profile, Business registration, Catalog management
- **seller-payouts** • Automated seller settlements, Escrow releases, Commission deductions
- **reviews** • Verified purchase product reviews, Ratings, Image feedback
- **coupons** • Promotional discount codes, Cart rules, Minimum order value constraints

Pillar 5 Farmer VIP Memberships & Plans — 4 Modules

- **farmer-plans** • VIP Tier plans (Free, Silver, Gold, Platinum), Prorated Upgrade Quotes
- **farmer-plan-payments** • Credit adjustments on upgrade, Online/UTR checkout, Downgrade lock
- **doctor-changes** • Advisor switch queue, Admin review workflow, 1-click Approval/Rejection
- **gardener-plans** • Urban gardening & home agri advisory plans and services

Pillar 6 Mandi Trading & King Connect B2B — 7 Modules

- **mandi-prices** • Live APMC commodity price feeds, State/District price trends
- **crop-sales** • Farmer crop sell listings, Buyer bidding, Minimum reserve price
- **j-forms** • Statutory APMC J-Form receipt generation, Weight receipts, Digital records
- **king-connect** • B2B direct trader-farmer connect, Bulk inquiries, Private contracts
- **b2b-demands** • Institutional buyer demand broadcasts, Procurement lot matching
- **king-payments** • B2B high-value escrow settlements, Milestone release conditions
- **p2p-ledger** • Shared double-entry ledger sync, Buyer-seller transaction reconciliation

Pillar 7 Wallets, Banking & Financial Ledger — 6 Modules

`wallet` • Digital cash wallet, Escrow holds, Ledger balance integrity

`wallet-transactions` • Credit/Debit transaction history, Purpose tagging, Reference IDs

`withdrawals` • Bank payout requests, Admin approval workflow, Batch IMPS/NEFT

`bank-accounts` • IFSC validation, Penny drop verification, Beneficiary registry

`tax-gst-tcs` • Statutory 1% TCS calculation, GST compliance reports, GSTR-8 returns

`icici-banking` • Connected banking APIs, Direct payout webhooks, Statement reconciliation

Pillar 8 Telemetry, Governance & Administration — 13 Modules

`satellite` • Sentinel/Landsat NDVI imagery integration, Field health heatmaps

`weather` • Hyper-local weather telemetry, Rain forecasts, Severe weather alerts

`soil-sensors` • IoT soil moisture, Temperature, Electrical conductivity telemetry

`deliveries` • Logistics route planning, Driver dispatch, Proof-of-delivery OTP

`fleet-tracking` • Real-time GPS coordinate telemetry, Driver status, Vehicle registry

`community` • Farmer social discussion board, Crop advisories, Knowledge sharing

`trainer-network` • Master trainer certifications, Field workshops, Cluster governance

`notifications` • Push notifications, SMS OTP gateway, WhatsApp alerts, Email dispatch

`admin` • Core admin controller, Platform KPI aggregation, Executive metrics

`settings` • Platform configuration switches, Feature toggles, Maintenance modes

`audit-logs` • Timestamped mutation logs across all 103 tables with actor & diff tracking

`iso-governance` • ISO 9001 & ISO 27001 data compliance audits, Security check protocols

`reports` • Master A4 printable reports, Balance sheets, Sales dossiers

4. Frontend Architecture - Expo React Native Web Client

frontend/ (Port: 8081)

```
frontend/
├── app/                                # File-Based Expo Router Directory
│   ├── index.tsx                       # Executive Super-Premium Landing Page (Hero, Mandi, AI Doctor, ISO Seals)
│   ├── _layout.tsx                     # Root Provider Tree (Theme, Auth, Toast, Audio Assistant, Safe Area)
│   ├── (auth)/                         # Authentication Flow
│   │   ├── login.tsx                   # Mobile/Password Login with Quick OTP Option
│   │   ├── register.tsx                # New Farmer / Buyer Registration with Referral Code
│   │   ├── forgot-password.tsx         # Password Recovery with SMS OTP Verification
│   │   └── otp-verify.tsx              # 6-Digit OTP Verification Screen
│   ├── (user)/                         # Farmer & General User Portal
│   │   └── (tabs)/                     # Primary User Navigation Tabs (Desktop & Mobile Responsive)
│   │       ├── index.tsx               # Farmer Home (Live Weather, Farm Health, Quick Actions)
│   │       ├── crops.tsx               # Crop Cycles, Plots, Field Operations, Expenses & Harvest
│   │       ├── ai-doctor.tsx           # AI Disease Scanner, Symptom Analysis, Agronomist Consultations
│   │       ├── mandi.tsx               # Live Mandi Rates, Crop Sale Listings, Buyer Demand Matches
│   │       ├── memberships.tsx         # VIP Tier Plans, Prorated Upgrades, Downgrade Protection Lock
│   │       ├── wallet.tsx              # Digital Passbook, Add Money, Bank Withdrawal, Referral Earnings
│   │       └── profile.tsx             # User Details, KYC Documents, Farm Records, Language Switcher
│   ├── (partner)/                      # Specialized Partner Dashboards
│   │   ├── doctor/                     # Agronomy Doctor Consultation Queue, Patient History, Prescriptions
│   │   ├── seller/                     # Multi-Vendor Store Portal, Product Listing, Orders, Payouts
│   │   └── driver/                     # Logistics Consignment Delivery, Route Navigation, Delivery Proof
│   ├── agristore/                      # Multi-Vendor Marketplace (Catalog, Product Details, Cart, Checkout)
│   └── king-connect/                   # B2B Mandi Direct Trading, Demand Requests, Shared Ledgers
├── src/
│   ├── api/                            # Axios Interceptors & API Clients (auth, crops, orders, plans, wallet)
│   ├── components/                     # Reusable UI Tokens (Header, GlassCard, LaserButton, LanguageSelector, AudioAssistant)
│   ├── hooks/                          # State Hooks (useAuth, useCrops, useOrders, useFarmerPlans, useWallet)
│   ├── context/                        # React Context Providers (AuthContext, ThemeContext, CartContext)
│   └── constants/                      # Colors, Obsidian/Slate Palettes, Typography, Breakpoints
```

5. Admin Console Architecture - Executive Platform Governance

admin/ (Port: 8082)

```
admin/
├── app/
│   └── admin/                          # Admin Navigation Root with Protected SuperAdmin Guard
│       ├── _layout.tsx                 # Admin Shell with Collapsible Sidebar, Quick Stats, System Alarms
│       └── (tabs)/                     # Executive Management Consoles
│           ├── index.tsx               # Platform Analytics (GMV, Active Farmers, Revenue, System Telemetry)
│           ├── admin-orders.tsx        # Multi-Vendor Order Fulfillment, UTR Payment Verification
│           ├── farmers.tsx             # Farmer Directory, Land Verification, KYC Dossier Approvals
│           ├── admin-sellers.tsx       # Vendor Storefront Onboarding, Commission Rates, GSTR-8 Audit
│           ├── memberships.tsx         # Plan Subscriptions, Doctor Switch Review Queue (1-Click Approve/Reject)
│           ├── mandi.tsx               # Mandi Market Rates, APMC Integration Feeds, Price Corrections
│           ├── finance.tsx             # Financial Ledger, Escrow Balances, Batch IMPS/NEFT Payouts
│           └── new-settings.tsx        # Platform Settings, Payment Gateways & [System Logs] Audit Tab (103 Tables)
├── src/
│   ├── components/                     # Admin Components (SettingAuditLogsView, AdminHeader, KycModal, DataTable)
│   ├── hooks/                          # Admin Hooks (useAdminOrders, useAdminFarmers, useDoctorChangeRequests, useAuditLogs)
│   └── api/                            # Admin API Interceptors & Endpoints
```

Core Architectural Compliance: 100% Pure React Native Web • Primary PC/Website View • 100% Pure English UI • Strict A4 Master Standard • Zero Live Auto-Deploy.

FARMSKING ENTERPRISE